

- (b) *Icinga Web*: This is also known as the new Web and has a Web 2.0 inspired front-end to offer drag and drop customised dashboards. It communicates to the core, database and other third party add-ons.
- 4. *Icinga Data Out Database*: This acts as a storage point for historical data monitoring for add-ons.
- 5. *Icinga Reporting*: This is a reporting module based on the open source Jasper Reports. The reporting module provides template based reports with varied access levels, and automated report generation and distribution.
- 6. *Icinga Mobile*: This is a user interface for smartphones and tablets. It is available for iOS, Android, BlackBerry, etc, and is based on JavaScript and Sencha Touch. Important plugins of Icinga are:
 1. Performance monitoring: PNP4Nagios, NagiosGrapher and InGraph
 2. Configuration interfaces and tools: Nconf, Nagios QL and LConf
 3. Business process monitoring: Business process add-ons
 4. Network visualisation: NagVis and Nagmap
 5. Windows monitoring: NSClient++ and Cygwin
 6. SNMP trap monitoring: SNMPTT and NagTrap

Op5 Monitor

Op5 is free and open source server and network monitoring software based on Nagios. Op5 specialises in displaying the status, health and performance of IT networks and has an integrated log server and Op5 logger. Op5 is developed and supported by Op5 AB.

The various products under Op5 are listed below.

1. *Op5 Free*: This is a perfect product for small IT offices. Basically, it is very easy to use and understand, and can monitor all types of servers and network devices, along with applications.
2. *Op5 Pro*: This is more suitable for organisations in need of single system development. It provides comprehensive monitoring for servers, network devices, applications, databases, storage and even cloud based services.
3. *Op5 Ent+*: This is suitable for large enterprises for monitoring devices and all sorts of servers.
4. *Op5 Live*: This is easy to use software available and suitable for everyone.

The following are the features of the Op5 monitoring software.

1. *Server monitoring*: Monitors all sorts of servers and provides alerts, reports and graph based visualisation. Op5 is efficient in monitoring physical, virtual, cloud and even hybrid server environments.
2. *Virtual monitoring*: Fully efficient network monitoring software for monitoring VMware ESX, vSphere, KVM, Citrix Xen and even Microsoft Hyper V.
3. *Cloud monitoring*: Op5 provides facilities to systems administrators to completely monitor SaaS, PaaS and IaaS, along with other types of cloud infrastructure.
4. *Open source*: As it is completely open source and

based on Nagios, there are no problems as such, in implementation.

5. *Scalable*: Op5 is highly flexible and scalable for monitoring large volumes of disk drives and handles distributed monitoring as well as load sharing in an easy manner.
6. *Data centre monitoring*: It is very efficient in managing and monitoring data centres comprising physical and virtual servers, application management and unified computing.
7. *Reporting*: It manages loads of information from various IT hardware and software, and presents the reports in a comprehensive manner in easy GUI based graphs for thorough understanding by systems administrators.
8. *Integrated log server monitoring*: The Op5 logger provides centralised storage to log various events, which enhances security and data integrity.

Important extensions of Op5 Monitor are:

- Op5 Monitor Peer
- Op5 Monitor Poller
- Op5 Monitor Cloud Extension

Munin

According to its official website, “Munin is a networked resource monitoring tool that can help analyse resource trends and “what just happened to kill our performance?” problems. It is designed to be very plug and play. A default installation provides a lot of graphs with almost no work.”

Munin is a free and open source network and system monitoring tool, which provides systems administrators with the great advantage of monitoring and alerting services for servers, switches, applications, etc. It is written in PERL programming language and uses the RRDTool to create graphs. It can be accessed via a simple Web interface. Munin provides comprehensive performance monitoring of computers, networks, SANs, applications, etc.

The latest version of Munin is 2.99.3 and its features are listed below:

- Munin runs a munin-node service on every monitored box, and the Munin server connects to the munin-node via TCP port 4949 to retrieve the data.
- Provides comprehensive data visualisation using graphs, giving the status as *OK*, *WARN*, *CRITICAL* or *UNKNOWN*.
- More than 500 monitoring plug-ins are available till date.

Network Management Information System

Network Management Information System (NMIS) is regarded as an open source network management system licensed under GNU license v3. It can play a crucial role in monitoring the performance of an organisation by measuring IT environments, assets and fault monitoring as well as other valuable information.

NMIS provides a highly scalable, flexible and easy to implement and maintain network monitoring environment for IT organisations. It can run both in physical and virtual environments, and can manage thousands of devices that have a vast amount of storage at a single point of time.